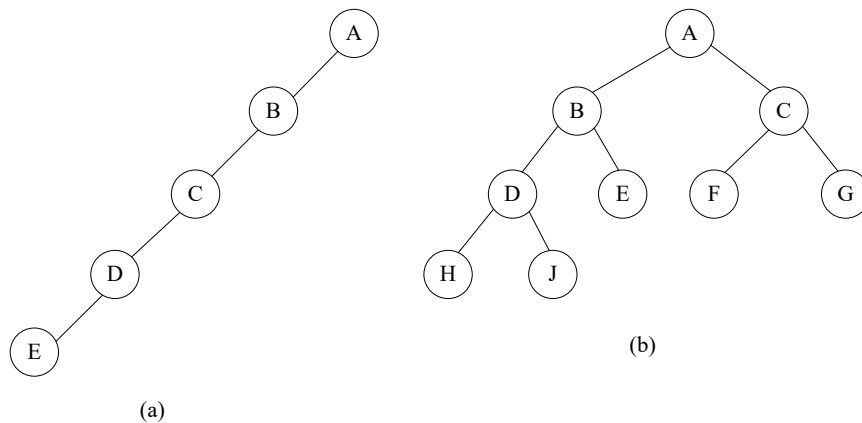
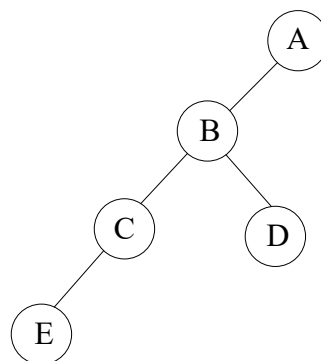


Homework 3.

- The file name of your homework (in PDF) should be in the format: “學號-作業編號.pdf”. For example: **00357999-ds-hw3.pdf**
 - Please submit your homework to Tronclass before **2025/12/21 23:59**.
1. (10%) What is the maximum number of nodes in a k -ary tree of height h ? Prove your answer.
 2. (20%) Write out the inorder, preorder, postorder, and level-order travels for the binary trees (a) and (b).



3. (15%) Considering operator precedence, draw the expression tree of the infix expression. Note that $^$ is power.
$$a / b - c ^ d * e ^ f ^ g + (h + i) / j$$
4. (10%) Draw the threaded representation of the following binary tree.



5. (10%) Suppose that we execute the following instructions:

**push 7, push 9, push 16, push 30, push 49, pop, push 82, push 5, push 33,
push 31, push 6, pop, push 2, push 1.**

- (a) What is the resultant heap if we use a max heap?
- (b) What is the resultant heap if we use a min heap?

6. (10%) Given the following numbers:

90, 60, 70, 100, 10, 50, 40, 20, 30, 80.

- (a) Construct a binary search tree.
- (b) From (a), what does the binary search tree become if we delete 90 using the rule “replaced by the largest element in its left subtree”?

7. (12%) How many distinct 6-node binary tree structures can we create?

8. (13%) Transforming the forest into a binary tree using the rule:

- If $T_1, \dots, T_n$ is a forest of trees, then the binary tree corresponding to this forest, denoted by $B(T_1, \dots, T_n)$
- Root equal to root(T_1)
- Left subtree equal to $B(T_{11}, T_{12}, \dots, T_{1m})$, where $T_{11}, \dots, T_{1m}$ are the subtrees of root (T_1); and has right subtree $B(T_2, \dots, T_n)$

